

House Price Prediction

Ames Housing Dataset - Full Project Report

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Kaggle Competition: House Prices - Advanced Regression Techniques

Key Results

Linear Regression CV RMSE: 0.1289 | Ridge CV RMSE: 0.1289

Target threshold: < 0.16 | Status: **PASSED**

Features used: 38 | Training rows: 1,458 | Test rows: 1,459

Part 1 - Exploratory Data Analysis

1.1 Numerical Features After Filtering

The first step was to filter the dataset to retain only columns with numerical data types (int64 and float64), discarding all categorical and object columns. After removing the Id column (a row identifier with no predictive value) and SalePrice (the target), we are left with 36 features.

S/n	Feature	#	Feature	#	Feature	Type
1	MSSubClass	13	BsmtFinSF2	25	GarageYrBlt	Mixed
2	LotFrontage	14	BsmtUnfSF	26	GarageCars	Count
3	LotArea	15	TotalBsmtSF	27	GarageArea	Area (sqft)
4	OverallQual	16	1stFlrSF	28	WoodDeckSF	Area (sqft)
5	OverallCond	17	2ndFlrSF	29	OpenPorchSF	Area (sqft)
6	YearBuilt	18	LowQualFinSF	30	EnclosedPorch	Area (sqft)
7	YearRemodAdd	19	GrLivArea	31	3SsnPorch	Area (sqft)
8	MasVnrArea	20	BsmtFullBath	32	ScreenPorch	Area (sqft)
9	BsmtFinSF1	21	BsmtHalfBath	33	PoolArea	Area (sqft)
10	BsmtFinSF2	22	FullBath	34	MiscVal	Dollar
11	TotalBsmtSF	23	HalfBath	35	MoSold	Month
12	BsmtFinSF1	24	BedroomAbvGr	36	YrSold	Year

1.2 Top 5 Features by Correlation with SalePrice

Pearson correlation was computed between each numerical feature and SalePrice. The five strongest positive correlations are shown below. The accompanying heatmap also reveals internal multicollinearity between predictors, which we address in Part 2.

Feature	Correlation (r)	Interpretation
OverallQual	0.7910	Strong positive - quality rating is the single best predictor
GrLivArea	0.7086	Larger living areas command higher prices

Feature	Correlation (r)	Interpretation
GarageCars	0.6404	Garage capacity proxy for property size/value
GarageArea	0.6234	Continuous version of garage capacity
TotalBsmtSF	0.6136	Basement square footage reflects total usable space

Note: GarageCars and GarageArea have an internal correlation of 0.887, flagging a multicollinearity issue addressed in Part 2.

1.3 Scatter Plot Analysis: Top 3 Features

OverallQual vs SalePrice: Near-monotonic positive relationship - higher quality consistently predicts higher price. The spread widens significantly at quality ratings 8-10, suggesting heteroscedasticity. The relationship is ordinal rather than perfectly linear, meaning equal steps in quality do not produce equal steps in price.

GrLivArea vs SalePrice: Broadly linear positive trend. Two extreme outliers appear at GrLivArea > 4,000 sqft with prices below \$200k - implausible for the Ames market given their OverallQual = 10 ratings. These are treated as partial or distressed sales and removed (see Section 1.5).

TotalBsmtSF vs SalePrice: Broadly linear with increasing variance at higher values. Houses with TotalBsmtSF = 0 (no basement) form a visible discrete sub-group at the left edge, pulling the regression line away from the main cluster.

1.4 SalePrice Distribution and the Case for Log Transformation

SalePrice is right-skewed with a skewness coefficient of 1.88. A long right tail means expensive homes disproportionately inflate the mean and distort OLS coefficient estimates, because the model penalises large errors at the high end more than proportionally.

Metric	Raw SalePrice	log(SalePrice)
Skewness	1.8829	0.1214
Mean	\$180,921	12.024
Std Dev	\$79,443	0.399
Min	\$34,900	10.460
Max	\$755,000	13.534

Applying log(SalePrice) reduces skewness from 1.88 to 0.12, producing a near-normal distribution. This satisfies OLS assumptions, stabilises residual variance (reducing heteroscedasticity), and aligns with the Kaggle competition scoring metric (RMSE on log scale). All modelling from Part 3 onwards uses log(SalePrice) as the target.

1.5 Outlier Identification and Decisions

Three distinct outlier groups were identified:

Id(s)	Description	Evidence	Decision	Justification
524, 1299	Very large homes, very low prices	GrLivArea >4,000 sqft; OverallQual=10; SalePrice <\$200k	DROP	Likely distressed/partial sales; would severely distort GrLivArea slope
Various	High-price luxury homes (>\$700k)	Follow the normal price-quality trend	KEEP	Legitimate luxury sales; removing would introduce selection bias
Various	Very low-price homes (<\$40k)	SaleCondition = Abnorml in many cases	KEEP	Plausible salvage/condemned sales; important variance to preserve

Part 2 - Data Cleaning

2.1 Missing Value Treatment

Only three numerical features had missing values. Each was treated with a strategy specific to the semantic meaning of the column - a blanket fillna(0) would be semantically wrong for features where zero already has a distinct meaning.

Feature	Missing	Strategy	Justification
LotFrontage	259 (17.7%)	Median imputation	LotFrontage is a continuous measurement with a right-skewed distribution; the median (69 ft) is more robust than the mean to extreme lot sizes. The high missingness rate (~18%) means dropping these rows would significantly reduce training data.
MasVnrArea	8 (<1%)	Fill with 0	A missing masonry veneer area almost certainly means no masonry veneer exists on the property. Zero square feet is the correct semantic value, not an unknown measurement.
GarageYrBlt	81 (5.5%)	Fill with 0	Cross-referencing GarageType for the same 81 rows confirms GarageType = NA (no garage). A missing build year here means there is no garage, not an unknown year. Filling with 0 cleanly separates the 'no garage' group from any real build year.

2.2 Multicollinearity Detection and Removal

Pairs of features with absolute Pearson correlation above 0.80 were identified from the correlation matrix. High inter-feature correlation inflates the variance of OLS coefficient estimates (the variance inflation factor, VIF), making individual coefficients unreliable and reducing model interpretability.

Feature A	Feature B	Correlation	Action	Feature Dropped
GarageArea	GarageCars	0.887	Drop GarageCars	GarageCars
GrLivArea	TotRmsAbvGrd	0.829	Drop TotRmsAbvGrd	TotRmsAbvGrd
1stFlrSF	TotalBsmtSF	0.804	Keep both	Neither

GarageArea was retained over GarageCars because it is a continuous, more precise measurement (square feet vs. a rounded integer car count). GrLivArea was retained over TotRmsAbvGrd because square footage is a direct physical measurement; room count is correlated but cruder and already captured by bedroom/bathroom counts. The 1stFlrSF / TotalBsmtSF pair was retained because they measure different floors of the house and each retains independent variance.

2.3 Feature Scaling

All features were standardised using StandardScaler (zero mean, unit variance) after feature engineering. Scaling matters for two reasons:

- Comparability: Raw features span very different numerical ranges (LotArea: 0-215,000 sqft vs. FullBath: 0-3). Without scaling, OLS coefficients are not directly comparable.
- Regularisation fairness: Ridge regression penalises the sum of squared coefficients. Without scaling, features with large raw units automatically receive larger penalties, meaning regularisation would be applied unequally - not because of predictive power, but because of unit choice. StandardScaler ensures Ridge penalises all features on the same footing.

Part 3 - Feature Engineering

3.1 Engineered Features

Four new numerical features were created by combining existing columns. Each is motivated by domain logic and was verified to improve correlation with SalePrice over its raw components alone.

Feature Name	Formula	Domain Logic	Corr. w/ SalePrice
TotalSF	1stFlrSF + 2ndFlrSF + TotalBsmtSF	Single best proxy for overall house size	0.782
HouseAge	YrSold - YearBuilt	Captures depreciation and renovation demand	-0.523

Feature Name	Formula	Domain Logic	Corr. w/ SalePrice
TotalBathrooms	FullBath + 0.5*HalfBath + BsmtFullBath + 0.5*BsmthalfBath	Unifies all bathroom measures with standard half-bath weighting	0.632
YearsSinceRemodel	YrSold - YearRemodAdd	A recently remodelled home commands a premium over an identical but dated one	-0.507

TotalSF achieves a correlation of 0.782 with SalePrice - higher than any of its three component features in isolation (1stFlrSF: 0.606, 2ndFlrSF: 0.319, TotalBsmtSF: 0.614). This confirms that the combined feature captures more signal than its parts.

3.2 Log Transformation of SalePrice

log(SalePrice) was used as the model target throughout Parts 3 and 4. The justification is threefold:

- Normality: Log transformation reduces skewness from 1.88 to 0.12, satisfying the OLS assumption of approximately normal residuals.
- Heteroscedasticity: The raw scatter plots show widening variance at higher prices. On the log scale, residual variance is more uniform across the price range.
- Percentage interpretation: On the log scale, coefficients can be interpreted as approximate percentage changes in SalePrice per unit increase in a feature — more economically meaningful than dollar-per-sqft for a skewed distribution.
- Competition alignment: The Kaggle scoring metric is RMSE on log(SalePrice), so training on the log target directly optimises the evaluation metric.

Part 4 - Modeling and Validation

4.1 Baseline: Linear Regression with 5-Fold Cross-Validation

A plain OLS linear regression was trained on all 38 scaled features (34 original numerical, after multicollinearity pruning, plus 4 engineered). A single train/test split was deliberately avoided because it produces unreliable generalisation estimates - the result depends heavily on which rows happen to end up in the test set. Five-fold cross-validation uses all rows for both training and evaluation, giving a more stable estimate.

Fold	RMSE (log SalePrice)	Notes
Fold 1	0.12154	
Fold 2	0.12823	
Fold 3	0.13814	Slightly higher - may include harder-to-predict rows
Fold 4	0.13019	
Fold 5	0.12660	
Mean	0.12894	Well below the 0.16 target threshold
Std Dev	0.00586	Low variance - stable across folds

4.2 Residual Analysis

After fitting the model on the full training set, the residuals (actual minus predicted log price) were examined. Key findings:

- The residual distribution is approximately normal and centred near zero, consistent with OLS assumptions being reasonably met.
- Slight heteroscedasticity remains at the high end of the predicted range - very expensive homes have larger absolute residuals.
- The five rows with the largest absolute residuals are homes where physical characteristics alone do not explain the sale price: one is a likely distressed sale (high quality, low price), others appear to be premium neighbourhood effects or timing anomalies not captured by numerical features.

The five largest prediction errors in the training set share a common characteristic: they are anomalous sales where the SaleCondition (e.g., Partial, Abnorml) or neighbourhood effects override what the physical characteristics would predict. This points directly to the biggest weakness of a numerical-only model.

4.3 Bonus: Ridge Regression

Ridge regression (L2 regularisation) was tested as an alternative to plain OLS. The regularisation hyperparameter alpha was selected automatically via cross-validation from the grid [0.01, 0.1, 1, 10, 50, 100, 200, 500].

Model	CV RMSE (log)	Std Dev	Notes
Linear Regression (OLS)	0.12894	0.00586	Baseline
Ridge (alpha = 50)	0.12895	0.00572	Marginally lower variance

What Ridge does (in plain English): Ordinary least squares minimises prediction error alone. Ridge adds a second penalty: keep the coefficients small. The hyperparameter alpha controls the trade-off — high alpha means more shrinkage. This prevents any single feature from dominating the model through an inflated coefficient, which is especially helpful when features are correlated.

Did Ridge improve the score? In this case, Ridge and OLS produce essentially identical results (difference < 0.00001 RMSE). This is expected: with 38 features and 1,458 training rows, the feature-to-row ratio is comfortable and overfitting is not severe. Ridge's benefit would be more pronounced with hundreds of features or stronger multicollinearity. The optimal alpha of 50 confirms mild regularisation was beneficial.

Competition submission: Final predictions used the Ridge model (alpha = 50), trained on all 1,458 training rows and predicting on the 1,459 test rows. Predictions were back-transformed from log scale using `exp()`.

Part 5 - Written Reflection

5.1 Hardest Decision

The hardest decision was how to handle the 81 missing values in `GarageYrBlt`. A missing year could mean one of two things: the property has no garage (fill with 0), or the garage exists but the build year was not recorded (fill with median year). These two interpretations lead to very different imputed values - 0 versus approximately 1978 - and choosing wrongly would silently corrupt a feature used in further calculations.

The resolution came from cross-referencing with the categorical `GarageType` column (not used in modelling but consulted during cleaning). All 81 rows with missing `GarageYrBlt` had `GarageType` = NA, confirming the 'no garage' interpretation. This is a good example of why domain understanding and cross-feature validation matter more than any mechanical imputation rule. Filling with 0 was correct here - but it would have been wrong if even a single row had a garage with a missing year. The lesson: always verify missing value patterns against related columns before imputing.

5.2 What the Model Gets Wrong

The five training rows with the largest absolute prediction errors reveal a clear pattern: the model mispredicts homes where the sale context overrides the physical characteristics. Specifically:

- High-quality homes sold at unusually low prices (likely foreclosures, estate sales, or family transactions below market rate) - the model sees `OverallQual` = 8-10 and predicts a premium price, but the actual `SaleCondition` drove the price down.
- Modest homes in premium neighbourhoods - the model has no neighbourhood signal (dropped as categorical), so it cannot distinguish a small house in Northridge Heights from an identical one in Brookside.
- Homes with unique features (elevators, tennis courts, second garages) - the `MiscVal` feature captures dollar value but the model cannot fully price unusual amenities.

In all cases, the model is making reasonable predictions given only physical measurements - the prediction errors reveal the information cost of dropping categorical features, not a flaw in the modelling approach per se.

5.3 Next Steps with One More Week

Priority 1: Encode Categorical Features

Neighbourhood alone explains enormous price variance that no amount of physical measurement can capture. One-hot encoding the 25 Ames neighbourhoods would likely reduce RMSE by 0.02-0.04 on its own. Ordinal encoding of quality ratings (`KitchenQual`, `ExterQual`, `BsmtQual`) would also add signal, since these have natural orderings that OHE does not preserve.

Priority 2: Log-Transform Skewed Input Features

Several input features are highly right-skewed - `LotArea` has a skewness of ~12, `GrLivArea` ~1.4. Applying `log1p()` transforms to these features would linearise their relationship with `log(SalePrice)`, potentially improving linear model fit without adding any new parameters.

Priority 3: Gradient Boosted Trees

XGBoost or LightGBM routinely achieve $RMSE < 0.11$ on this dataset. These models handle non-linear interactions, categorical features, and missing values natively without the manual engineering required for linear regression. They would serve as a strong benchmark to measure the ceiling of what is achievable with this data.

Priority 4: Stacking / Ensembling

A simple average of predictions from Linear Regression, Ridge, and a tree-based model typically outperforms any individual model, because the models make different errors on different rows. Even a two-model average of Ridge + XGBoost would likely push RMSE below 0.11.

Final Model Performance

Linear Regression CV RMSE: 0.1289 | Ridge CV RMSE: 0.1289

SUCCESS: Both models pass the < 0.16 RMSE threshold
